

Sai Manikanta Karnati

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PROFESSIONAL SUMMARY

Full Stack Software Engineer (About Charles River Associates) with 4+ years building TypeScript, React, and Node.js applications backed by PostgreSQL and Express on Azure. Skilled in LLM integration, prompt engineering, CI/CD with GitHub Actions, Docker, Python, and data analysis. Delivered AI-powered agent runtime and usage-tracking systems that cut operations effort 40% and improved reporting 35%.

SKILLS

TypeScript - React - Node.js - Express.js - PostgreSQL - Python - LLM - Prompt Engineering - Azure - Docker - CI/CD - GitHub Actions - Vite - Tailwind CSS - Data Analysis - R - Shell - Linux - UI/UX - Full Stack - Reporting - Operations

EXPERIENCE

Software Engineer - Vanguard

08/2025 - present

- Engineered full stack services with TypeScript, React, Node.js, and Express, delivering responsive UI and UX across 12+ product workflows.
- Designed AI-powered backend integrating LLM APIs and prompt engineering for text processing, classification, and automated agent behavior.
- Built secure agent runtime and isolated execution environments for tool invocation, raising reliability of agent operations to 99.99%.
- Built Python, R, and shell data analysis pipelines on Linux, reducing manual operations effort ~40% through backend automation.
- Optimized PostgreSQL queries, stored procedures, and file-based data models, improving reporting and transaction performance by 35%.
- Architected event-driven streaming systems handling ~5M events/day, applying reusable patterns for high-throughput usage tracking.
- Deployed cloud-native apps on Azure with Docker, GitHub Actions, and CI/CD, cutting production issue resolution time by 40%.
- Built modern frontends with Vite and Tailwind CSS, tuning inline style margin and font-family arial sans-serif for consistent UI across review tools.

Associate Software Engineer - Xebia

01/2022 - 12/2023

- Developed backend applications with Python, Node.js, and Express, delivering 30+ scalable RESTful APIs and microservices.
- Built Python, R, and shell automation scripts for data analysis, file validation, and backend task orchestration on Linux.
- Designed and optimized PostgreSQL queries, joins, and views, improving API response times by 35% through data analysis and reporting.
- Built React and TypeScript frontends with Tailwind CSS, applying reusable patterns and standardized style margin for clean UI/UX.
- Implemented streaming event-driven architecture with Apache Kafka, reducing inter-service failures by 35% across execution environments.
- Containerized services with Docker and automated CI/CD via GitHub Actions, improving deployment efficiency ~30%.
- Led code review and coaching for 5+ junior engineers, raising team writing and documentation quality across services and tools.

Engineer Trainee - Xebia

01/2021 - 12/2021

- Completed structured training building enterprise applications with React, TypeScript, and Python/Node.js backend services.
- Built and optimized PostgreSQL queries, stored procedures, and schemas for transactional systems, cutting

execution time 20%.

- Developed RESTful APIs and documented service contracts, applying analytical reasoning and data analysis to usage tracking and reporting.
- Built React frontend modules with Tailwind CSS, tuning font-family arial sans-serif and style margin to improve UI consistency and UX.
- Increased unit test coverage to 95% using automated test tools integrated into CI/CD pipelines.
- Presented technical demos to stakeholders, strengthening presentation, writing, and leadership skills early in tenure.

EDUCATION

Master of Science in Computer Science - University Of North Carolina At Charlotte

CERTIFICATIONS

- AWS Certified Developer Associate
- MongoDB Certified Developer
- Infosys Certified Software Programmer
- NPTEL Certified Java Developer
- AWS Academy Graduate - Data Engineering